HealthFirst Care – Operational Data Analysis Report

1. Introduction

This report presents the data cleaning, manipulation, and analysis conducted on operational datasets from HealthFirst Care. The objective was to identify trends in appointment scheduling, patient satisfaction, and resource utilization to support workflow optimization and improve patient experience.

The datasets analyzed include:

- Appointment Data
- Patient Feedback Data
- Resource Utilization Data

The analysis was performed using Microsoft Excel, leveraging data cleaning techniques, lookup functions, Pivot Tables, slicers, and visualizations.

2. Data Cleaning and Preparation

The following data preparation steps were completed:

Removed duplicate records across all datasets

Standardized date formats to MM/DD/YYYY

Standardized time formats to HH:MM AM/PM

Replaced missing numerical values (e.g., Feedback Score) with column averages

Replaced missing non-numerical values with “N/A” where appropriate

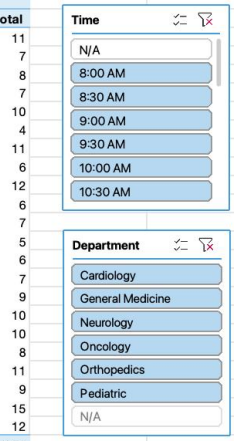
Linked datasets using INDEX-MATCH based on Patient ID

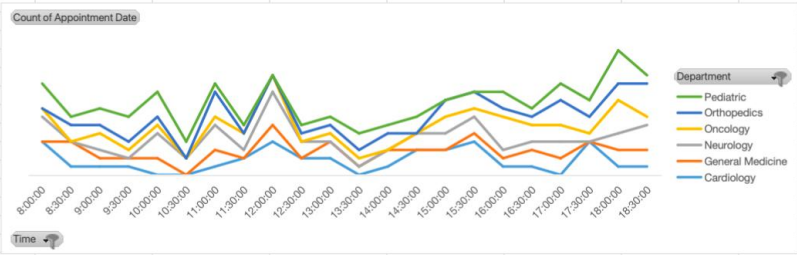
A combined dataset was created to enable cross-analysis of appointment details and patient satisfaction metrics.

3. Appointment Trends Analysis

Peak Appointment Hours

Count of Appointment Date	Department						
Time	Cardiology	General Medicine	Neurology	Oncology	Orthopedics	Pediatric	Grand Total
8:00:00	4		3	1		3	11
8:30:00	1	3			2	1	7
9:00:00	1	1	1	2	1	2	8
9:30:00	1	1		1	1	3	7
10:00:00		2	3	1	1	3	10
10:30:00			2			2	4
11:00:00	1	2	3	1	3	1	11
11:30:00	2		1	2		1	6
12:00:00	4	2	4	2			12
12:30:00	2		2		1	1	6
13:00:00	2	2		1	1	1	7
13:30:00		1		1	1	2	5
14:00:00	1	2			2	1	6
14:30:00	3		2			2	7
15:00:00	3		2	2	2		9
15:30:00	4	1	2	1	2		10
16:00:00	1	1	1	4	1	2	10
16:30:00	1	2	1	2	1	1	8
17:00:00		2	2	2	3	2	11
17:30:00	4			1	2	2	9
18:00:00	1	2	2	4	2	4	15
18:30:00	1	2	3	1	4	1	12
Grand Total	**37**	**26**	**34**	**29**	**30**	**35**	**191**





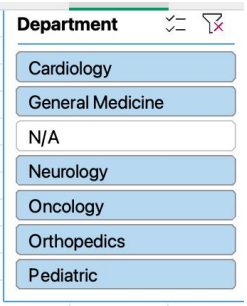
Pivot Table analysis of appointment counts by department and time revealed:

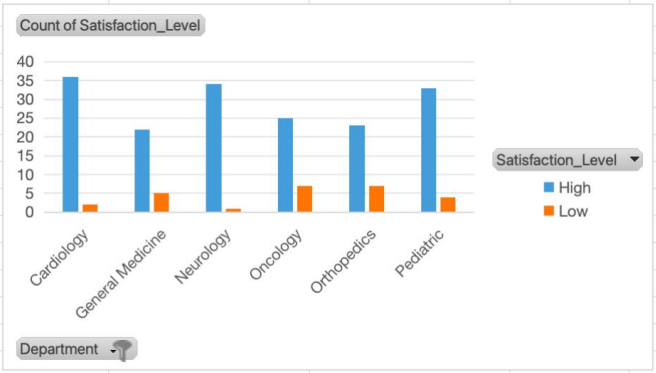
- Higher appointment volumes during mid-morning (around 11:00 AM–12:00 PM)
- Another visible spike during late afternoon (around 5:30 PM–6:30 PM)
- Pediatric and Orthopedics departments showed consistently higher appointment volumes

These peak time windows indicate operational pressure points where patient wait times may increase and staff workload intensifies.

4. Patient Satisfaction Analysis

Count of Satisfaction_Level	Satisfaction_Level		
Department	High	Low	Grand Total
Cardiology	36	2	38
General Medicine	22	5	27
Neurology	34	1	35
Oncology	25	7	32
Orthopedics	23	7	30
Pediatric	33	4	37
Grand Total	**173**	**26**	**199**





Feedback data was categorized into:

1. High Satisfaction (Score ≥ 4)
2. Low Satisfaction (Score < 4)

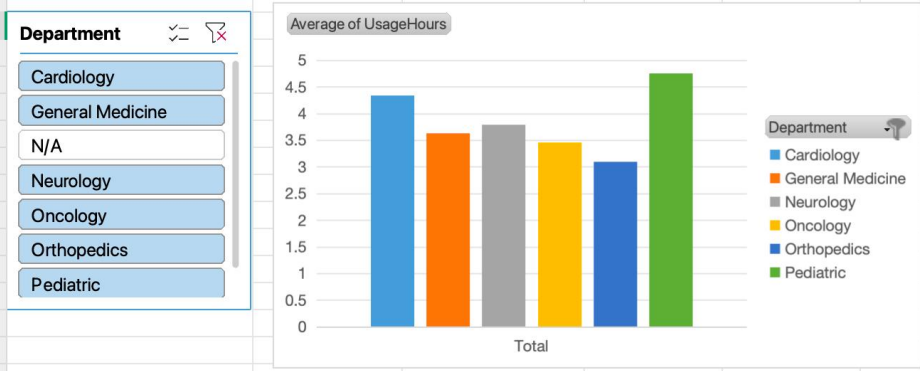
Findings include:

- Overall satisfaction levels are predominantly high
- Oncology and Orthopedics show relatively higher counts of low satisfaction scores compared to other departments
- Comments such as “Delayed response” and “Long wait times” correlate with peak appointment hours

This suggests that service delays during high-demand periods may negatively impact patient experience.

5. Resource Utilization Analysis

	Department					
	Cardiology	General Medicine	Neurology	Oncology	Orthopedics	Pediatric
Average of UsageHours	4.347826087	3.634146341	3.794117647	3.461538462	3.09375	4.761904762



Average usage hours were calculated using Pivot Tables across departments.

Key observations:

- Pediatric department shows the highest average usage hours
- Orthopedics reflects comparatively lower utilization
- Some departments demonstrate moderate utilization despite high appointment volumes

This imbalance suggests potential inefficiencies in resource allocation and scheduling alignment.

6. Key Findings

- Peak workload occurs during late morning and late afternoon
- Pediatric and Orthopedics handle significant patient volumes
- Dissatisfaction appears linked to wait times and response delays
- Resource utilization varies across departments, indicating potential redistribution opportunities

7. Recommendations

Based on the analysis:

- Reallocate staffing during peak hours to reduce wait times
- Review scheduling density during high-demand time slots
- Optimize resource distribution across departments
- Implement proactive patient communication during peak periods

These improvements can enhance patient satisfaction while improving operational efficiency.

8. Conclusion

Through structured data cleaning and analytical techniques in Excel, meaningful insights were derived from HealthFirst Care's operational datasets.

The findings highlight actionable opportunities to optimize scheduling, improve patient experience, and enhance resource planning. These insights can support informed decision-making and drive continuous operational improvement.